

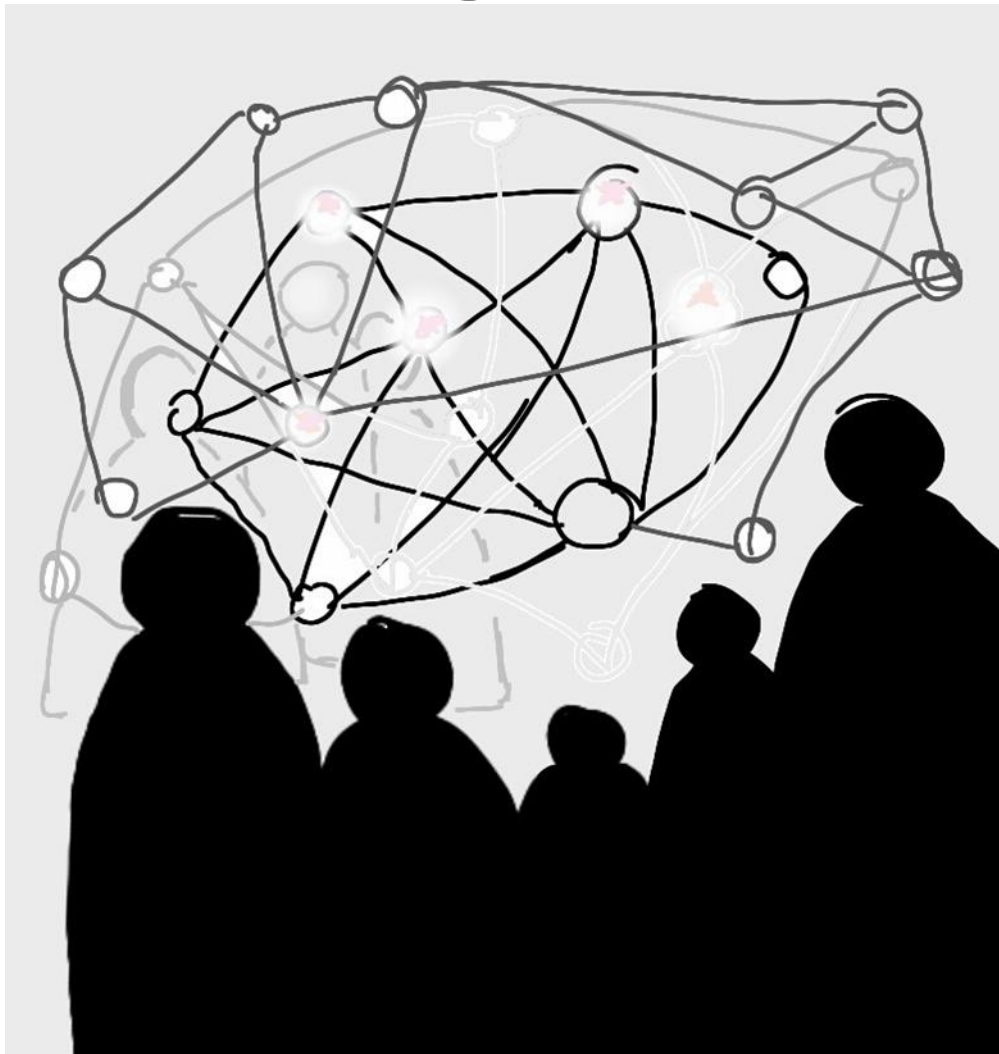
AI 507: Artificial Intelligence and Society



#14 EntertAInment

Associate Prof. Dr. Lena Frischlich

Flashlight



Flashlight

Health & wellbeing

Definition of Health

Salutogenese

Coherence

Well-being

Life satisfaction

Satisfaction of intrinsic needs
(mastery, autonomy & affiliation)

Hedonic

Eudaimonic

Psychologically rich

AI for wellbeing

AI negative consequences for wellbeing

Delusion

Despair

Deaths

Small share – but still many
people

Sycophancy – validation -
violence

AI and psychotherapy

Faster diagnosis (e.g., of depression/
anxiety)

Reduced waiting times

Certified online apps

Therapeutic support of
specialised models

But: Fairness in different
subgroups matters

AI and medicine

Great potentials

Personalized medicine

Image interpretation

Multimodal data

Approval by central organisations
& insurance companies

Great challenges

Personalized medicine

Data access

Accountability

Great opportunities for further
development

**Do you have questions to
last week?**



Then let's get started!

After today's lecture, you will hopefully have...

- ❑ ... a better understanding of why people play
- ❑ ... learned how people perceive virtual worlds and fiction
- ❑ ... discussed how virtual realities affect our empathy
- ❑ ... gotten first hand experiences into how one of Copenhagen's best ranked start-ups <https://modl.ai/> uses AI to make games ready for their future fans

**Why do we talk about
games?**

Games are the worlds largest entertainment industry

(Accenture, 2021; <https://gamesdenmark.dk/industry-in-numbers>; Danske Statistik, 2018)

**\$300
Billions**

Estimated value of
the industry –
more than film &
music combined



50%



52%



<https://www.deviantart.com/miloscat/art/Butterfile-s-986651824>

Between 2017- 2022:
74% more full-time
employees in the
gaming industry

**3,6 mio.
DKK**
Revenue
2022

Why do people play?

Uses & Gratifications

(Katz, Blumler & Gurevitch, 1973)

Key question: Why do people turn to media?

Basic assumptions

1. People have social and psychological needs
2. They have expectations of mass media (and other sources) regarding the satisfaction of these needs
3. This leads to different usage patterns
4. These patterns of use lead to the satisfaction of needs (gratification) – but can also have unintended consequences

Central: Focus on the user – not the normative evaluation of the use



Elihu Katz via
[/www.asc.upenn.edu](http://www.asc.upenn.edu)



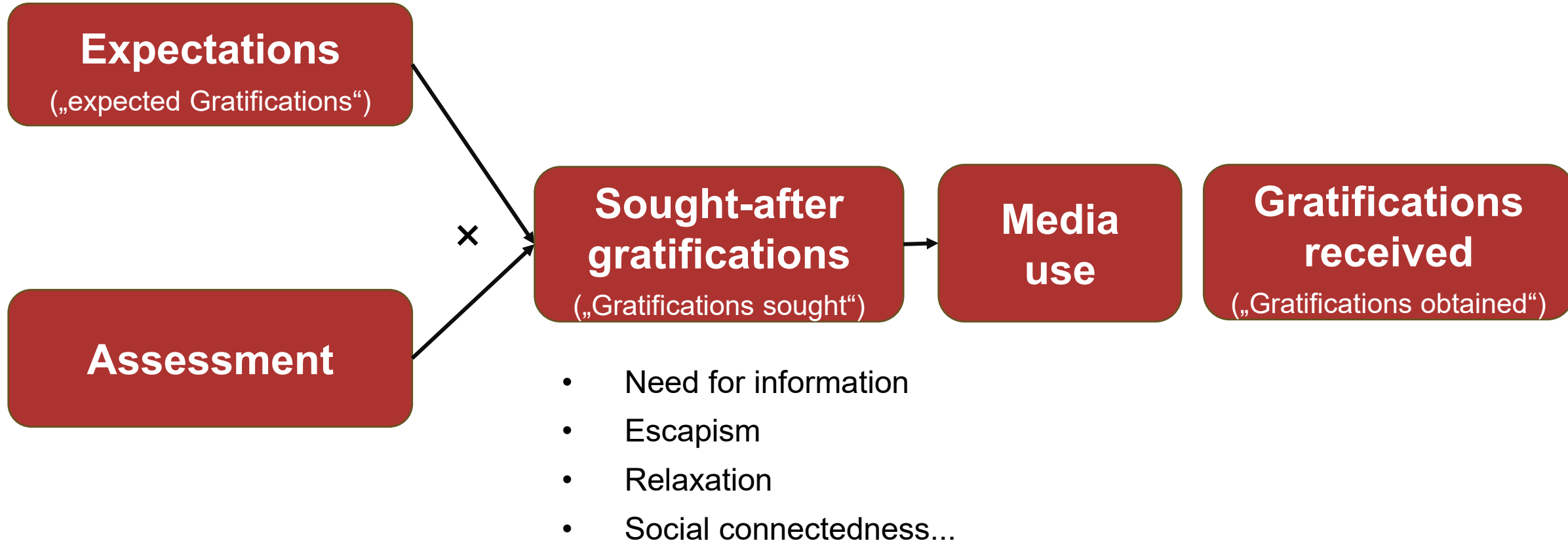
Jay Blumler via University of
Maryland



Michael Gurevitch via
<http://teoriausosygratificacions.blogspot.com/2010/05/los-teoricos.html>

Or, as a process

(Katz, Blumler & Gurevitch, 1973)



Gaming specific motives (Chea et al., 2022; Sailer et al., 2017; Tang et al., 2020; Yee, 2006)

N = 3000 Massive Multi-Player Online Role Game (MMPoRG) Player

Factor analysis approach

Three key factors

- **Achievements**
- **Social affairs,**
- **Immersion**

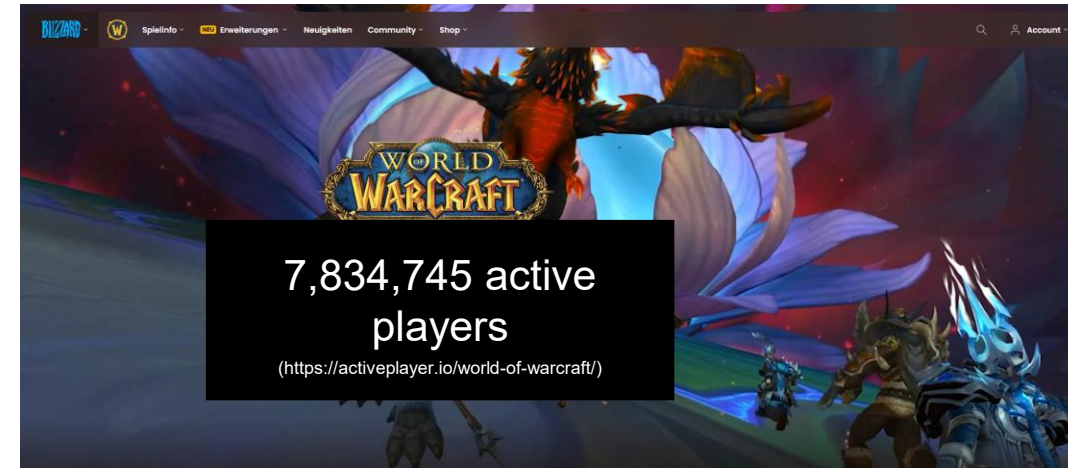
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- **Gamification** = application of elements from computer games in other contexts
- In particular, features such as badges, leaderboards, and performance graphics satisfy needs for mastery
- Central for e-sports
- Game addiction is more associated with escapist and social motives – goal orientation is a rather „healthy play“

2) Social Interactions/ connectedness

(Bowman, Rieger & Lin, 2022; Domahidi, Festl & Quandt, 2014; Kowert, Domahidi, & Quandt, 2014)

- Gaming is (also) a place for friendships
- Especially those who play a lot and have a strong social motive to play are more likely to make lasting connections online
- Gaming can even help emotionally insecure people to form social relationships
- Can contribute to psychological well-being
- Social experiences are part of the attraction of many games (and game adjacent platforms such as discord, twitch or steam)



<https://worldofwarcraft.blizzard.com/de-de/>

S

3) Identification

(Bessi re et al. 2007; Chea et al., 2022; Guadagno et al., 2011; Qi et al., 2023)

Avatars as a means of self-exploration

- Often represent the ideal self ("how you would like to be") rather than the real self ("how you are")
- Opportunity to try out roles (e.g., in violent or gendered games)
- Doesn't mean that avatars necessarily overcome stereotypes...
- Players identify with their avatars and can experience them as part of their self-concept, up to a more automated processing of the avatar compared to other familiar content

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<https://na.leagueoflegends.com/en/featured/events/avatar-creator>

digital democracy centre

4) Immersion and flow

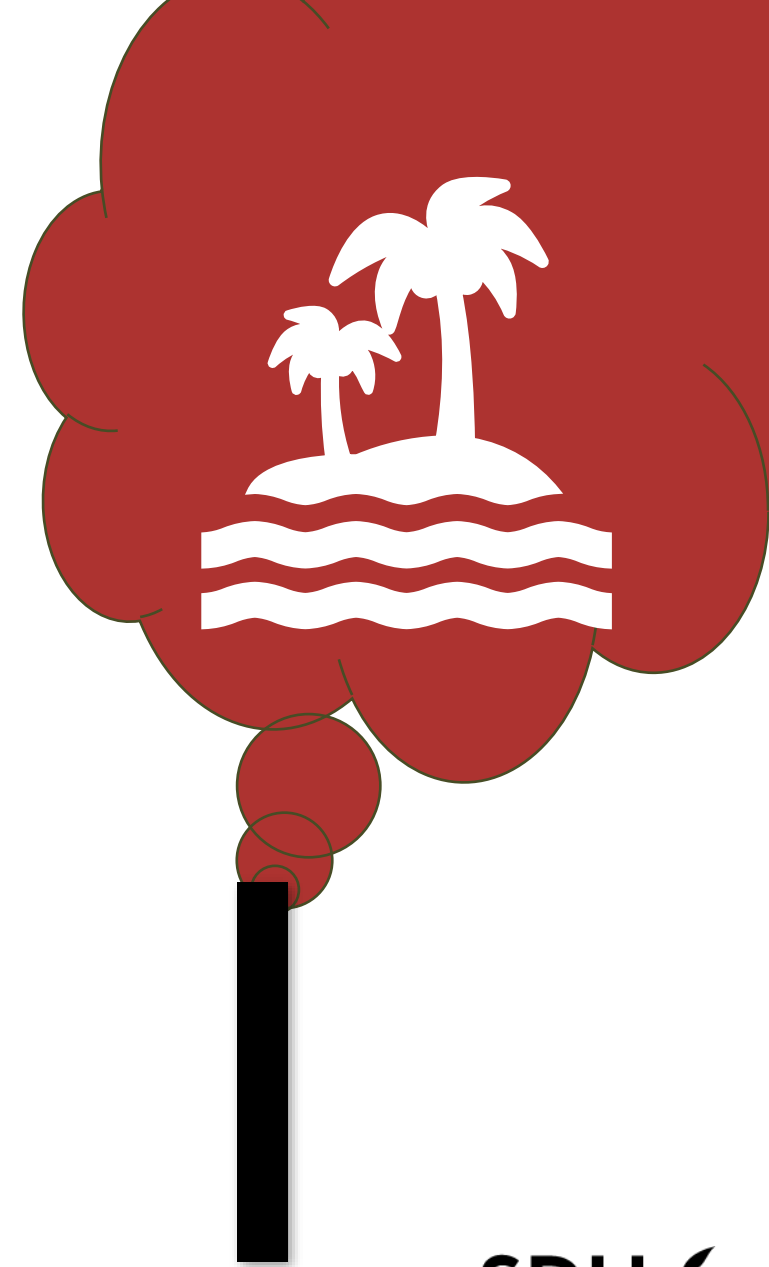
(Csikszentmihalyi, 2014; Michailidis et al., 2018; Therrien, 2023)

- **Immersion** = immersion in the virtual environment, credibility, liveliness, sensuality of experience – an imperfect, gradual state
- **Presence** = The feeling of being "there", of "getting lost" in the digital environment; the feeling "that this is real"
- **Flow** = Holistic state, total involvement
 - Action after action results according to an internal logic
 - We merge with our environment, stimulus and reaction, past, present and future become one



(5) Escapism (Hussain et al., 2021)

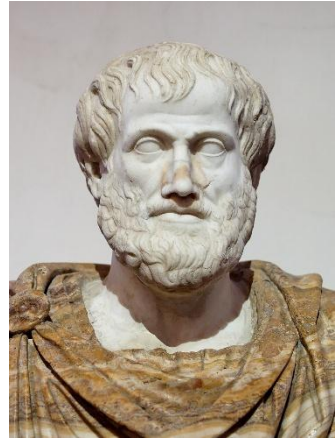
- Immersion to distract yourself from everyday problems
- Negative correlates: depression, social phobias, loneliness, etc.
- Positive correlates: e.g. positive mood, wishes, joy
- But meta-analysis of 27 studies ($N = 28.893$) showed that the association with negative aspects is more pronounced, including addictive behavior (same we also saw in problematic AI use last week)
- Effect more pronounced in Western (i.e., compared to Eastern) cultures
- More cultural comparative research needed!



Content factors: Entertainment processes

(Bartsch et al. 2016; Oliver et al. 2018; Oliver & Raney, 2011; Tamborini et al. 2011; Vorderer & Reinecke, 2015; Wirth et al. 2012; Zillmann, 1988)

- *Hedonic* entertainment (e.g., via comedies)
 - Characterised by feelings of fun, pleasure, or suspense
 - *Central for mood management*
- a more complex, *eudaimonic* entertainment (e.g., via dramatic movies, meaningful, value-laden content)
 - characterized by mixed and often self-transcending emotions such as nostalgia, meaningfulness or appreciation
 - “thought-provoking” entertainment experiences
 - can inspire self-transcending experiences and motivate pro-social behavior



Aristoteles; Lysipp - Jastrow (2006), via <https://commons.wikimedia.org/w/index.php?curid=1359807>

Content factors: Narrative processing

(Cohen et al., 2015; de Graaf et al., 2012; Green & Brock, 2000)

1. Transportation

- the feeling of getting lost in the world of a narrative, immersing yourself and leaving the real world behind (→ pretty similar to immersion/ presence)
- integrated fusion of attention, images, and emotions

2. Identification

- Changing the self-concept by adopting perceived characteristics of a media person
 - Understanding, taking perspective and empathy
- VR enhances the transportation and identification through *being* and *acting* in *there* (aka presence)

Questions



Augmented & Virtual Reality

Augmented Reality

(Azuma, 1997; Devarigi et al., 2022; LaViola et al., 2017)

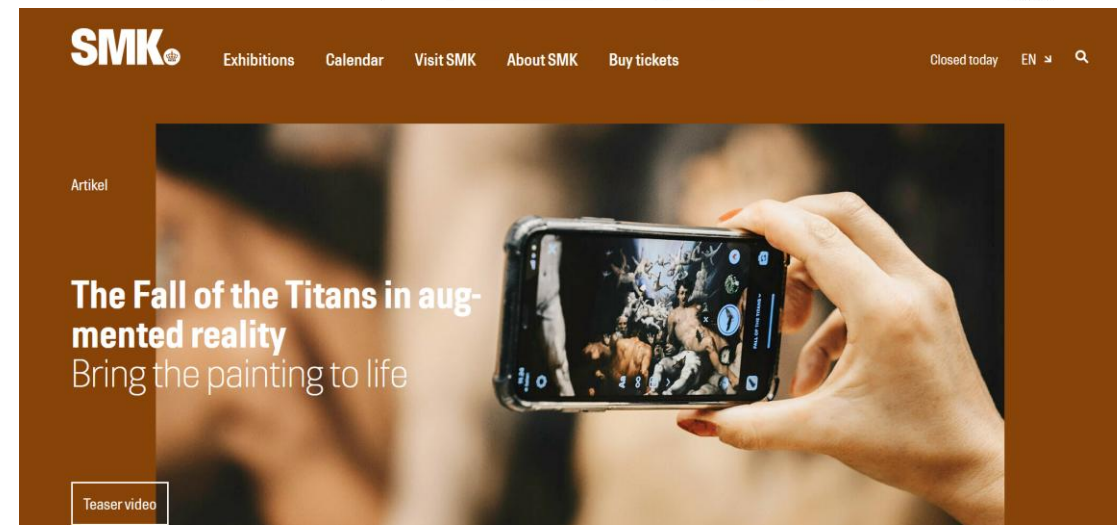
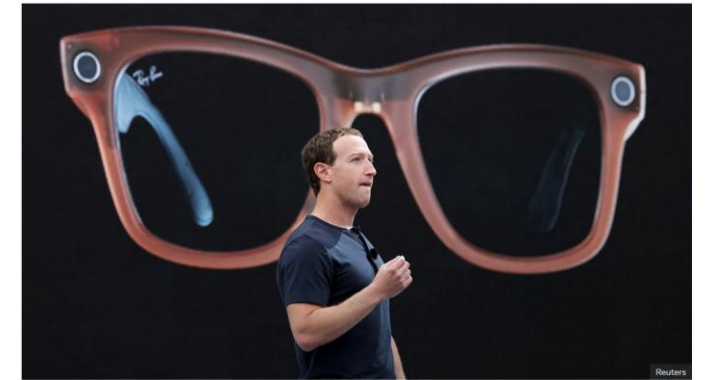
- Approaches to extend (= augmentation) the user's view of a real environment through synthetic or simulated information
- Augmented reality systems typically combine the real and the virtual, are interactive in real time, and are captured in 3D
- Examples
 - 1) Augmented reality technology based on mobile phones/tablets, or
 - 2) Applications with optically transparent or video-transparent head-mounted displays
- E.g., Meta's Rayban collaboration

Smart glasses are 'an invasion of privacy' - Meta's are selling better than ever

5 days ago

Share Save Add as preferred on Google

Kali Hays
Technology reporter



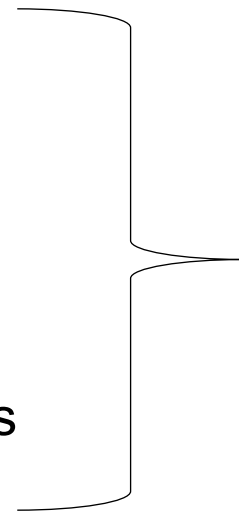
<https://www.smk.dk/en/article/titanernes-fald-med-augmented-reality/>

Virtual Reality

(Biocca, 1992; Fox et al., 2009)

Interactive Media Systems

- Record users' movements
- Responsively create a rich sensory environment
- which is intended to replace the cues from the physical environment



- a wider range of sensory impressions (e.g. visual representations, movement, depth, sound, spatiality, balance)
- Naturally depicted interaction possibilities through head rotation, gestures and body movements
- Higher level of interactivity
- Therefore, more immersive

VR as an empathy machine? (Herrera et al., 2018)

Suggested in a TED Talk by filmmaker Chris Milk, in which he argues that VR can be the ultimate "empathy machine"

But is that really the case??

Experiment 1

- a) Idea of what it would be like to be homeless vs. b) VR, in which you are homeless in the 1st person perspective
- *N = 113 participated in follow-up surveys for 8 weeks*
- More empathy in VR short term, but not in the long run
- No differences in attitudes but more signatures on a petition 8 weeks later

Experiment 2 (N = 452) showed that VR worked better than a less immersive medium (desktop) and replicated the results



Fig 2. Scene progression for VR perspective-taking task. Top: Participant's point of view in the apartment scene. Middle: Participant's point of view in the car scene. Bottom: Participant's point of view in the bus scene.

Embodiment & the Proteus Effect

(Bem, 1972; Banakou et al., 2016; Fox et al., 2013, Hoter & Nagar, 2023; Kilteni et al., 2013; Pena et al. 2009; Ratan et al. 2020; Yee & Bailenson, 2007, Yee et al., 2009)

- Similar idea to the SIDE-Model (→ Recall #4)
- But: Here avatars serve as the “salient” identity
- Remember: Our self-concept is based on what we do (working self-concept → Recall #4)
- When users are "embodied", they accept the virtual body as their own and behave accordingly
- Users align their attitudes and behavior with the characteristics they attribute to the avatar= Proteus-Effect

Activates stereotypes

- Better drum performance with Jamaican avatars
- More self-objectification as women (vs. men)

Reduce stereotypes

- Less stereotypes after playing a Black avatar
- Less prejudice and more engagement for disabled people



Fig. 6

From: The effects of a wheelchair simulation in a virtual world



View of a participant at the post office

Time for a break







modl.ai

Building the future of AI-driven tools for game development.

Computerspiele · Copenhagen V, Capital · 4 Tsd. Follower:innen · 11-50 Beschäftigte

Niels Jørgen & 1 weiterer Kontakt folgen dieser Seite



Then I hope you now have

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Wrap up – all things multiple choice exam

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